Prodigy Anesthesia Quick Review

Adult Respiratory Distress Syndrome

What is ARDS?	Adult Respiratory Distress Syndrome (also known as Acute Respiratory Distress Syndrome) which is defined as acute hypoxemic respiratory failure due to inflammatory pulmonary injury. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 205.*
What are the first signs of ARDS?	The first sign of ARDS is arterial hypoxemia that is resistant to treatment. The chest xray will be indistinguishable from cardiogenic pulmonary edema. Increased airway resistance, reduced functional residual capacity, and increased intrapulmonary shunt also result from ARDS. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 205.*
What is the pathogenesis of ARDS?	The acute phase of ARDS is due to an accumulation of protein-filled fluid into the alveoli due to increased pulmonary capillary permeability. The inflammatory response that results in increased capillary permeability is mediated by neutrophils and cytokines within the lungs. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 205.*
What disorder has the highest risk of progression from acute lung injury to ARDS?	Sepsis *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 205.*
How is ARDS diagnosed?	For a patient to meet the diagnosis of Adult Respiratory Distress Syndrome (also known as Acute Respiratory Distress Syndrome), four criteria must be met: acute onset, PAO2 to FiO2 ratio <200 regardless of the level of PEEP applied, bilateral infiltrates on chest xray, and a PA wedge pressure less than or equal to 18 mmHg. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 205.*
How is the pulmonary capillary wedge pressure affected by ARDS?	The PCWP is typically normal in these patients. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 205.*
How does right-sided heart failure result from ARDS?	Pulmonary vasoconstriction due to hypoxemia results in pulmonary hypertension that can cause right-sided heart failure if severe enough. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 205.*

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What are the current recommendations for the use of high tidal volumes and PEEP in the treatment of patients with ARDS?

Currently, high tidal volumes (12-15 ml/kg) are known to be associated with decreased pulmonary compliance and the risk for barotrauma. It is recommended to adjust ventilator settings so that peak inspiratory pressures do not exceed 35-40 cm H2O. PEEP is extremely useful in these patients as it helps prevent alveolar collapse, increases functional residual capacity, decreases right-to-left shunting, and improves V:Q matching.

Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 206-207.

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